IE 643: Deep Learning - Theory and Practice **Aug-Nov 2024**

Project End-term Report : General video to anime video translation

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Abstract

The demand for high-quality anime-style transformations of real-world visual media has increased with the popularity of anime-inspired aesthetics in various digital art forms. To address the challenge of achieving realistic, fine-grained anime stylization for both static images and video sequences, this work presents a three-stage pipeline called Scenimefy. This approach begins with a pseudo paired data generation stage, where a real-to-anime domain transfer is facilitated by fine-tuning a StyleGAN model. This process produces pseudo paired datasets that help align semantic content between real and anime domains. Next, a segmentation-guided data selection step refines these datasets by ensuring coherence and relevance of stylized textures. In the final stage, a semi-supervised image-to-image translation framework is applied, incorporating both supervised and unsupervised training branches to optimize for high-fidelity, anime-style detail. In particular, the supervised branch utilizes a patch-wise contrastive style loss to ensure local texture accuracy, while the unsupervised branch captures broader stylistic characteristics from a large set of real and anime images. By combining these elements, Scenimefy achieves realistic and context-aware anime stylization in both images and video sequences, paving the way for high-quality, stylized animations in anime aesthetics.

Github link : `https://github.com/rubul27/Scenimefy.git`

1 Introduction

Transferring real scene images into anime styles presents formidable challenges stemming from the complexity of the scenes, the unique features of anime, and the significant domain gap between the two. Real-world scenes often consist of multiple objects with intricate relationships and hierarchical arrangements between foreground and background elements, requiring a nuanced understanding of spatial relationships to maintain the integrity of the original scene during translation.

Anime's distinctive textures and intricate details—such as hand-drawn brush strokes in landscapes—demand a level of stylistic fidelity that goes beyond simpler forms typically seen in earlier studies. Replicating these artistic nuances while preserving the overall composition is a significant challenge. Additionally, the lack of high-quality datasets that accurately capture the anime aesthetic exacerbates the domain gap issue. Many existing datasets focus on human faces and foreground objects, leading to inconsistencies in style that adversely affect the overall quality of the translated images.

To address these challenges, we present Scenimefy, a novel three-stage framework designed for the seamless translation of real images into anime styles. This approach incorporates pseudo paired data generation, segmentation-guided data selection, and a semi-supervised image-to-image translation strategy. By leveraging a patch-wise contrastive loss to enhance local spatial details, we aim to preserve the intricate textures and relationships present in both real and anime scenes. Ultimately, our method offers a robust solution that not only mitigates the challenges posed by complex scene structures and unique anime characteristics but also enhances the quality and aesthetic appeal of the resulting images.

2 Literature Survey

2.1 Image-to-Image Translation

Image-to-image translation is a core area of research aimed at transforming images from one domain to another. Notably, the Pix2Pix model introduced by Isola et al. in 2017 has become foundational in this field. By utilizing conditional Generative Adversarial Networks (GANs), Pix2Pix effectively learns to map input images to output images when paired training data is available. The model's ability to produce high-fidelity images while maintaining structural integrity is a key strength; however, its dependence on paired datasets limits its applicability in scenarios where such data is difficult to obtain.

In response to the challenges of acquiring paired data, several unsupervised models have emerged, including CycleGAN and the Contrastive Unpaired Translation model (CUT). These models typically operate under the cycle-consistency constraint, allowing them to learn transformations without requiring direct correspondences between images. Despite their advantages, such as flexibility in training, these approaches can struggle with semantic preservation in complex scenes, leading to artifacts in the output images.

2.2 Domain Adaptation of StyleGAN

The domain adaptation of StyleGAN has garnered significant attention, with various strategies developed to transfer knowledge from pre-trained models to new domains effectively. StyleGAN, known for its impressive image generation capabilities, operates by separating the content and style of images, allowing for diverse output generation. While this model excels in producing high-resolution images, it faces challenges when applied to image-to-image translation tasks, particularly regarding maintaining the semantic relationships within complex scenes.

Recent advancements in domain adaptation techniques, such as FreezeG and attribute-level adaptation via CLIP embedding spaces, aim to overcome some of these limitations. These strategies allow for more targeted fine-tuning of StyleGAN models for specific domains, enhancing their ability to generate images that adhere to desired stylistic constraints. However, limitations remain, such as the risk of overfitting and the challenges of modeling intricate scene relationships.

2.3 Scene Cartoonization

Cartoonization has emerged as a popular application of image-to-image translation, particularly in transforming real-world images into cartoon-like representations. CartoonGAN and its lightweight variant, AnimeGAN, represent notable contributions to this area. CartoonGAN employs a combination of adversarial loss functions to promote clear edges and smooth shading, effectively capturing the cartoon aesthetic. However, while it excels in global structural preservation, it often falls short in maintaining detailed local textures.

AnimeGAN extends these ideas specifically to the anime style, incorporating anime-specific loss functions to achieve appealing results while remaining computationally efficient. Despite their strengths, both CartoonGAN and AnimeGAN may produce outputs that lack depth and nuanced detail, which can detract from the overall visual quality. The focus on achieving a specific aesthetic can also limit their adaptability to varied artistic expressions.

Loss Functions in AnimeGAN

AnimeGAN employs a combination of loss functions to ensure effective image transformation from real-world photos to anime-style images, balancing the preservation of content and style throughout the generation

process. The key loss components are:

Adversarial Loss $L_{\mathbf{adv}}(G, D)$ Based on the Least Squares Generative Adversarial Network (LSGAN) framework, this loss function drives the generator G to produce images that are indistinguishable from real anime images, while the discriminator D aims to correctly classify real and generated images. This adversarial training encourages high-quality, realistic output from the generator.

Content Loss $L_{\mathbf{con}}(G, D)$ The content loss measures the difference between the generated image and the input content image in terms of high-level features. It is computed using a pre-trained VGG19 network to extract semantic features, ensuring that the generated images maintain the essential content of the original photographs. This helps to retain important visual details during the transformation.

Grayscale Style Loss $L_{\mathbf{gra}}(G, D)$ This loss encourages the generated images to adopt the stylistic elements of anime by calculating the Gram matrix of feature maps extracted from the VGG network. The grayscale style loss promotes the desired textures and line art characteristic of anime, although it can risk producing images with reduced color.

Color Reconstruction Loss $L_{\mathbf{col}}(G, D)$ To address the potential grayscale output from the grayscale style loss, the color reconstruction loss is incorporated. This loss ensures that the generated images not only adopt the style of anime but also retain the colors of the original content images, creating a harmonious blend of style and color fidelity.

The overall loss function for AnimeGAN is defined as:

$$L(G, D) = \omega_{\text{adv}} L_{\text{adv}}(G, D) + \omega_{\text{con}} L_{\text{con}}(G, D) + \omega_{\text{gra}} L_{\text{gra}}(G, D) + \omega_{\text{col}} L_{\text{col}}(G, D)$$

where $\omega_{\text{adv}}, \omega_{\text{con}}, \omega_{\text{gra}}$, and ω_{col} are weights that balance the contributions of each loss function.

3 Methods and Approaches

In this section, we present a comprehensive methodology designed to facilitate effective image-to-image translation from real scenes to anime styles, which we term Scenimefy. Our approach is structured into three main components, each addressing a crucial aspect of the translation process:

Pseudo Paired Data Generation: We leverage a pretrained StyleGAN model to create pseudo paired datasets that bridge the gap between real-world and anime imagery. By generating pairs of corresponding real and anime scenes, we ensure that the model learns the stylistic and structural relationships necessary for effective translation. This is achieved through a fine-tuning process that retains the foundational features of the original generator while adapting to the anime style.

Semantic Segmentation Guided Data Selection: To enhance the quality of the generated data, we employ a semantic segmentation model (Mask2Former) to filter the pseudo paired dataset. This filtering process ensures that only high-quality pairs with semantic consistency are retained. We focus on pixel-wise alignment and diversity of semantic categories to enrich the dataset, which is critical for training a robust translation model.

Semi-Supervised Image-to-Image Translation: Our training framework consists of both supervised and unsupervised branches. The supervised branch utilizes the pseudo paired data to train a conditional GAN, ensuring high-fidelity translations. Meanwhile, the unsupervised branch employs unpaired real and anime datasets to refine the model further, utilizing novel loss functions that promote consistency and differentiation among image patches. This dual approach allows the model to balance fidelity to the original content with the artistic elements of the anime style.

To mitigate the domain gap between real-world and anime scenes, we propose a novel approach to generating

pseudo paired data. This method involves utilizing a pretrained StyleGAN model, referred to as G_s, which is originally trained on a comprehensive dataset of real scenes. The aim is to adapt this model to generate corresponding anime scenes, leading to the formulation of a new generator G_t through fine-tuning on an anime dataset. This fine-tuning process is critical for aligning the stylistic attributes of the two domains. Following the successful application of image-to-image translation, we proceed to convert a real video into an anime-style video through a systematic process that includes three key steps: frame extraction, style transfer, and video reconstruction.

Frame Extraction: We begin by utilizing OpenCV to extract individual frames from the input video. Each frame is treated as a separate image, allowing us to apply our anime style transfer model independently to each frame.

Style Transfer: Next, each extracted frame undergoes processing through our trained anime style transfer model, such as Scenimefy or AnimeGAN. This step applies the desired anime aesthetic while preserving the original content and context of the scene.

Video Reconstruction: Finally, we compile the newly styled anime frames back into a cohesive video format. This reconstruction process ensures smooth playback and maintains a consistent visual style throughout the video.

This structured approach enables the effective transformation of live-action footage into a visually captivating anime representation, preserving the narrative flow and dynamic motion of the original video.

3.1 Pseudo Paired Data Generation Process

Latent Space Sampling: We begin by sampling latent codes w from a standard normal distribution $N(0, I)$. These codes are essential as they serve as the input to the generator, allowing us to create diverse outputs.

Generating Pairs: Using the latent codes w, we generate the following pairs:

$$x_p = G_s(w) \in X_p \quad \text{and} \quad y_p = G_t(w) \in Y_p$$

Here, x_p represents the generated pseudo real scene, while y_p denotes the corresponding anime scene. This process results in a set of pseudo paired data $\{x_p, y_p\}$.

Preservation of Structural Information: To maintain the structural integrity of the generated scenes, we implement a strategy of freezing the initial layers of the generator G_t during the fine-tuning phase. This ensures that the fundamental features learned from the real scenes are retained while allowing the later layers to adapt to the stylistic nuances of the anime dataset.

3.2 Loss Function for Fine-Tuning

To align the semantic and stylistic features of the generated scenes, we introduce a combined loss function that includes:

CLIP Loss: Measures the cosine distance between the CLIP embeddings of x_p and y_p:

$$D_{CLIP}(x_p, y_p) = 1 - \text{cosine}(f_{CLIP}(x_p), f_{CLIP}(y_p))$$

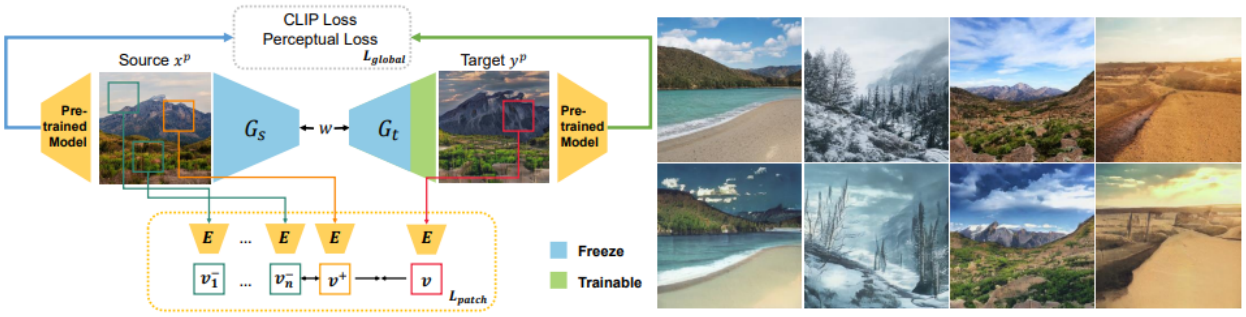


Figure 1: Overview of semantic-constrained fine-tuning strategy for pseudo paired data generation. Left: We initialize a source generator Gs and a target one Gt by pre-trained on a real scene domain. Gs remains fixed throughout the process. Gt is optimized using rich pre-trained model prior (i.e., CLIP, VGG) guidance with the early layers freezed to be adapted to an anime scene domain. A patch-wise contrastive loss using pre-trained CLIP embedders E is applied to better preserve local spatial details. Right: Examples of generated pseudo paired data after segmentation-guided data selection

Perceptual Loss: Computed using the LPIPS metric, which captures perceptual similarity by comparing deep features:

$$lpips(x_p, y_p) = \sum_{i=1}^{L} \frac{1}{N_i} \sum_{j=1}^{N_i} \|f_i(x_p) - f_i(y_p)\|_2^2$$

The overall global loss function for the fine-tuning stage is given by:

$$L_{global} = D_{CLIP}(x_p, y_p) + \lambda_{lpips} \cdot lpips(x_p, y_p)$$

where λ_{lpips} is a weight that balances the contribution of the perceptual loss.

Patch-wise Contrastive Loss: To enhance local details and style consistency, we introduce a patch-wise contrastive loss, L_{patch}. This loss is defined as:

$$L_{patch}(v, v^+, v^-) = -\log \left[\frac{\exp(v \cdot v^+)}{\exp(v \cdot v^+) + \sum_{i=1}^{N} \exp(v \cdot v_i^-)} \right]$$

where v is the feature vector of a patch, v^+ is a positive patch (same style), and v^- are negative patches (different styles).

The complete objective for the fine-tuning stage can then be summarized as:

$$L_{finetune} = L_{GAN_t}(G_t, D) + \lambda_{global} L_{global} + \lambda_{patch} L_{patch}$$

where L_{GAN_t} is the adversarial loss specific to the generator G_t and discriminator D.

3.3 Semantic Segmentation Guided Data Selection

Following the generation of pseudo paired data, we implement a robust filtering mechanism guided by semantic segmentation. This process employs the Mask2Former model to ensure the generated pairs maintain semantic integrity.

3.3.1 Data Filtering Criteria

Semantic Consistency: We calculate the pixel-wise cross-entropy loss L_{BCE} to quantify the semantic alignment between the generated pairs. Pairs that exceed a defined loss threshold (5.0) are discarded. This metric ensures that only those pairs with a high degree of semantic coherence are retained.

Semantic Abundance: We also evaluate the number of distinct categories detected in each image pair. Pairs with only one identifiable category are removed to enrich the dataset with diverse semantic information. This ensures that the training data contains enough variety to facilitate effective learning.

This data selection strategy ensures that the subsequent training phases are based on high-quality and semantically rich paired data.

3.4 Semi-Supervised Image-to-Image Translation

Our semi-supervised image-to-image translation framework is composed of two distinct training branches: the supervised branch and the unsupervised branch.

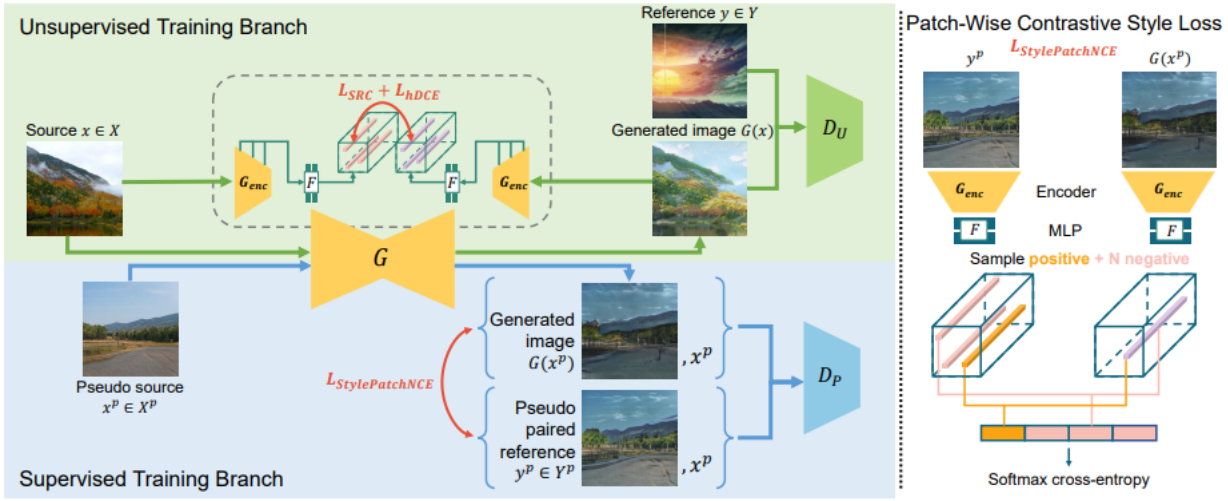


Figure 2: Main framework of semi-supervised image-to-image translation. Left: The proposed approach comprises two branches: unsupervised (top) and supervised (bottom). The supervised branch ingests the pseudo paired data as supervision, while the unsupervised branch learns the true target distribution using the original real and anime datasets. A novel patchwise contrastive style loss is proposed to learn local fine details better. Right: Details of the patch-wise contrastive style loss.

3.4.1 Supervised Training Branch

The supervised branch leverages the pseudo paired data to implement a conditional GAN framework, aiming to achieve high-fidelity translations from real scenes to anime styles.

Loss Functions in the Supervised Branch

Conditional GAN Loss: The adversarial loss function is defined as:

$$L_{cGAN}(G, D_P) = E_{y_p, x_p}\left[\log D_P(y_p \mid x_p)\right] + E_{x_p}\left[\log\left(1 - D_P(G(x_p) \mid x_p)\right)\right]$$

Style Patch NCE Loss: Our introduced StylePatchNCE loss promotes local style similarity across image patches:

$$L_{StylePatchNCE}(G, F, Y_p) = \sum_{l=1}^{L} \sum_{i \neq j} L_{patch}^{style}(v_i^l, v_i^l, v_j^l)$$

The total objective for the supervised branch can thus be expressed as:

$$L_{sup} = L_{cGAN}(G, D_P) + \lambda_{style} L_{StylePatchNCE}(G, F, Y_p)$$

where λ_{style} controls the contribution of the StylePatchNCE loss to the overall training objective.

3.5 Unsupervised Training Branch

The unsupervised branch directly ingests the original high-quality real dataset $\{x_i\}_{i=1}^{N}$ and the anime dataset $\{y_j\}_{j=1}^{M}$ to learn the true target domain distribution. Inspired by Jung et al. [reference16], we recognize the importance of addressing heterogeneous semantic relations of the image patches within a complex scene image. For instance, the patches from a mountain or a sea, and even their different parts, contain diverse semantic information. Such semantic relations should be considered and preserved for plausible unsupervised scene stylization.

Accordingly, we apply the semantic relation consistency loss L_{SRC} and the hard negative contrastive loss L_{hDCE} for training [reference16]. L_{SRC} minimizes the Jensen-Shannon Divergence (JSD) of the in-image patch similarity distribution between x and $G(x)$, enhancing semantic consistency during translation. L_{hDCE} implements the patch-wise contrastive loss that gradually increases the discriminative difficulty of negative samples, enhancing the model's discriminative power. For loss details, please refer to Jung et al. [reference16]. We apply both losses to the features of x and $G(x)$ extracted by G_{enc} and F, similar to our StylePatchNCE loss.

The total loss of the unsupervised branch is written as:

$$L_{unsup} = L_{GAN}(G, D_U) + \lambda_{SRC} L_{SRC} + \lambda_{hDCE} L_{hDCE}, \tag{1}$$

where L_{GAN} is the adversarial loss [reference10], and D_U is a standard discriminator for unsupervised training. λ_{SRC} and λ_{hDCE} are the loss weights.

3.6 Overall Training

The full framework of Scenimefy is thus semi-supervised, seeking a trade-off between faithfulness and fidelity of scene stylization. The complete loss function is defined as:

$$L_{i2i} = L_{unsup} + \lambda_{sup} L_{sup}, \tag{2}$$

where λ_{sup} decays gradually following a cosine function as the training progresses.

3.7 Work done before the prep-presentation review

Data Collection Curation: For initial experimentation, video frames were collected from real-world scenes and anime datasets. These frames served as foundational data for the style transfer models.

Preprocessing: The collected video frames underwent several preprocessing steps, including frame extraction, resizing to a uniform resolution, and normalization to standardize input quality. This preprocessing ensured consistency across both real-world and anime datasets, allowing for smoother training and evaluation.

Model Experimentation: To explore potential approaches for anime-style translation, I experimented with multiple models:
StyleGAN: Enabled fine-grained control over style and content, allowing for a more nuanced approach to anime-style rendering.
CycleGAN: A baseline for unpaired image-to-image translation.
CartoonGAN: Designed specifically for cartoon-style translation, providing insights into stylization methods.
AnimeGAN: A model tailored for anime style transfer, which helped set comparative benchmarks for anime-specific features.
These experiments informed further development and highlighted areas for enhancement, particularly in style fidelity and semantic consistency.

3.8 Work done after the prep-presentation review

After the prep-presentation review, we finalized our choice of Scenimefy as the model for real-to-anime style translation. Scenimefy was selected based on its ability to achieve a balance between anime style fidelity and content preservation, addressing limitations observed in other models like CycleGAN, CartoonGAN, and AnimeGAN.

Implementation and Fine-Tuning
To implement the Scenimefy model, we utilized a dual-branch approach that combines supervised and unsupervised learning to achieve high-quality anime-style transformations. The model relies on both real-world scene photos and anime images, processed through a semi-supervised image-to-image translation framework.

Supervised Training on Pseudo Paired Data
First, we generated pseudo-paired data by taking a pre-trained StyleGAN2 generator and fine-tuning it on our anime dataset. Using the same latent codes, we generated 30,000 image pairs by adjusting only the final few layers of the generator for anime-specific styling while preserving the structure. To enhance diversity without sacrificing quality, we applied a mild truncation trick with a threshold of 0.7, and a segmentation-guided data selection strategy to improve the relevance and consistency of the paired data.

Unsupervised Training Branch
For the unsupervised branch, we processed real-world and anime scene images independently. This branch aimed to enhance style consistency by learning patterns and textures unique to anime. The loss functions used in this branch included style and reconstruction losses to retain both the artistic qualities of anime and the semantic integrity of real-world scenes.

Loss Function Optimization
We applied a balanced set of hyperparameters: λ_{style} at 0.05, λ_{SRC} at 0.05, λ_{hDCE} at 0.1, and a gradually decreasing weight $\lambda_{\text{sup}(t)}$ based on cosine scheduling over 20 epochs. The entire model was trained on a single NVIDIA GeForce RTX 3090 GPU, ensuring efficient convergence and style adaptation.

This method enabled Scenimefy to capture anime-style elements, like texture and color balance, while preserving the content and structure of the original real-world frames. The fine-tuning process was carried out for 1,000 iterations to achieve optimal style transfer without overfitting to specific anime features, allowing the model to generalize effectively across a wide range of scenes.

Evaluation and Results
We assessed the effectiveness of Scenimefy using the Fréchet Inception Distance (FID) score, a quantitative metric to measure the quality of generated images. Scenimefy achieved a significantly lower FID score compared to baseline models, indicating closer alignment with anime-style characteristics and superior perceptual quality.

Application to Video Conversion
Building on these results, we applied the fine-tuned Scenimefy model to video content. By processing each frame of real-world videos individually, Scenimefy was able to produce continuous anime-style sequences with consistent style and smooth transitions. This final step demonstrated the model's robustness in real-time video-to-anime translation, effectively bridging the gap between static image translation and dynamic video transformation.

4 Dataset Details

4.1 Dataset

This method employs a semi-supervised image-to-image translation framework utilizing real-world scene photos and anime images for the unsupervised branch, complemented by a pseudo paired dataset for the supervised branch. All images are resized to 256×256. The training set comprises the dataset from and 6,656 scene images from CycleGAN for testing.

We also created a high-resolution (1080×1080) Shinkai-style anime dataset consisting of 5,958 images sourced from nine prominent Shinkai Makoto films, including *Weathering with You* (2019) and *Your Name* (2016). This dataset excludes irrelevant and low-quality images, mitigating overfitting and narrowing the domain gap. Also 1553 animation images from the movie "Paprika" directed by Kon Satoshi and 3000 animation images from the movie "SummerWar" are used.

Additionally, we generated a pseudo paired dataset of 30,000 images using the same latent codes from the source and fine-tuned StyleGAN generators, employing a mild truncation trick with a threshold of $\gamma = 0.7$ to enhance data quality.



Figure 3: Examples of our anime scene dataset. We showcase the image samples of various anime scenes.

5 Experiments

We begin by training a StyleGAN2 generator on the LHQ dataset at a resolution of 256×256. Subsequently, we fine-tune this generator on our curated anime dataset, allowing only the last three layers to be trainable while freezing the other layers. The hyperparameters used during this fine-tuning are $\lambda_{\text{lpips}} = 0.01$, $\lambda_{\text{global}} = 1.0$, and $\lambda_{\text{patch}} = 0.05$, and this process is carried out for 1,000 iterations.

Following the fine-tuning, we generate 30,000 pseudo paired data using a truncation threshold of 0.7, employing the proposed data selection scheme. The implementation of the unsupervised training branch is built upon a recent image translation model. We trained it for 20 epochs with hyperparameters $\lambda_{\text{style}} = 0.05$, $\lambda_{\text{SRC}} = 0.05$, $\lambda_{\text{hDCE}} = 0.1$, and $\lambda_{\text{sup}}(t) = \cos\left(\frac{\pi}{40}(t-1)\right)$, where $t = \{1, 2, \ldots, 20\}$ represents the current training epoch.

6 Results

Our **qualitative comparison** with CartoonGAN and AnimeGAN illustrates that Scenimefy strikes an optimal balance between style fidelity and semantic consistency. While CartoonGAN and AnimeGAN use handcrafted anime-specific losses to enhance edge sharpness, this approach limits their ability to achieve a high degree of stylization, often resulting in artifacts, particularly in backgrounds like the sky and sea. In contrast, Scenimefy effectively captures intricate anime features while maintaining semantic accuracy. For instance, our model successfully replicates the brush strokes of leaves, showcasing superior perceptual quality compared to the state-of-the-art methods.



Figure 4: Qualitative Comparison of Anime Style Transfer. Rows show results from AnimeGAN (top), Scenimefy (middle), and source images (bottom). Scenimefy captures finer anime details with improved style fidelity over AnimeGAN.

Method	LHQ (real)	CartoonGAN	AnimeGAN	Ours
FID↓	121.807	67.200	67.739	**58.104**

Figure 5: **Table 1:**Quantitative comparison using FID. A lower FID is better. The anime scene dataset is used as a reference.

Quantitative Results: Table 1 presents a quantitative assessment of our method in comparison to the baseline approaches. Our method achieves the lowest Fréchet Inception Distance (FID) score, which demonstrates that the quality of our translated results is superior and aligns with our observations of enhanced visual quality. Additionally, we evaluate the FID score between the real and anime scene datasets as a reference point. The style distribution of our translated results is significantly more aligned with the anime domain than with the real dataset.

7 Future Work

Model Optimization and Refinement:

Continue refining the Scenimefy model by exploring advanced training techniques, including data augmentation strategies and hyperparameter tuning based on previous evaluation outcomes. Investigate the potential for incorporating additional datasets to enhance the model's versatility, allowing it to adapt to various styles and improve performance across diverse inputs.

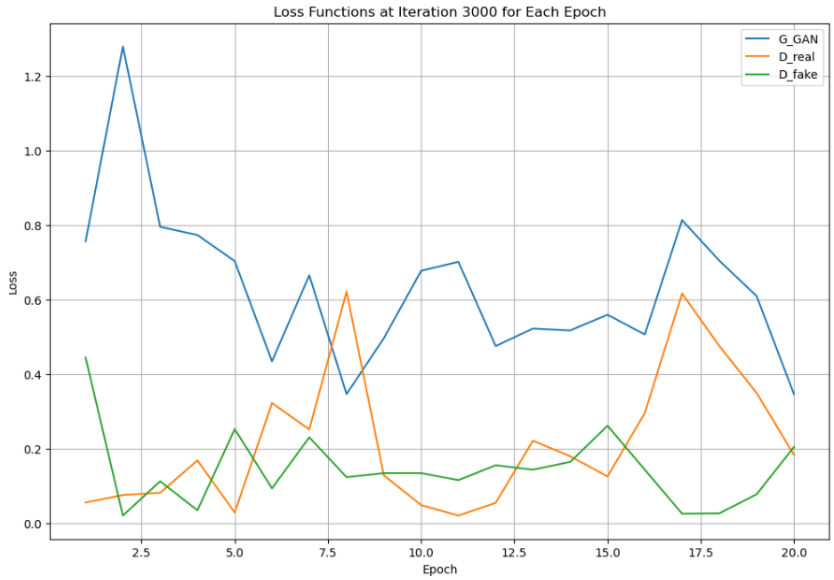


Figure 6: Loss functions : $\mathrm{G}_{GAN}, D_{real} D_{fake}$

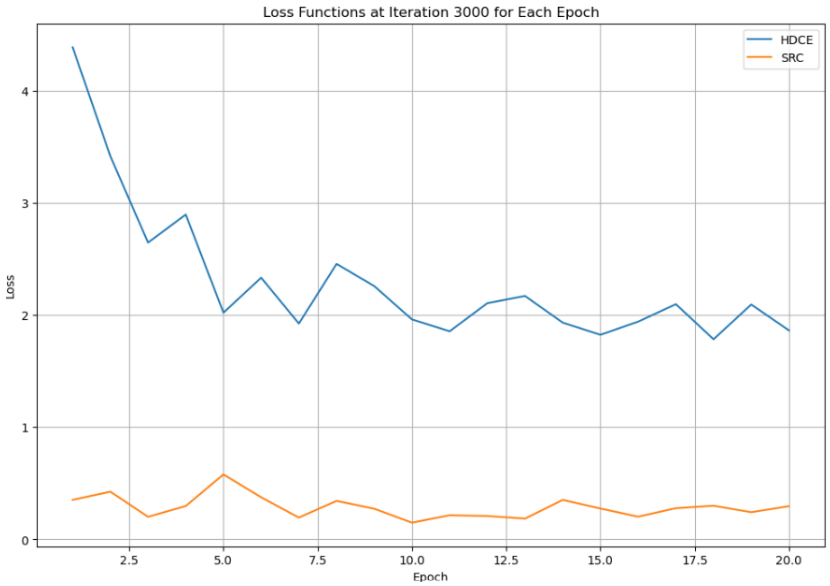


Figure 7: SRC HDCE loss

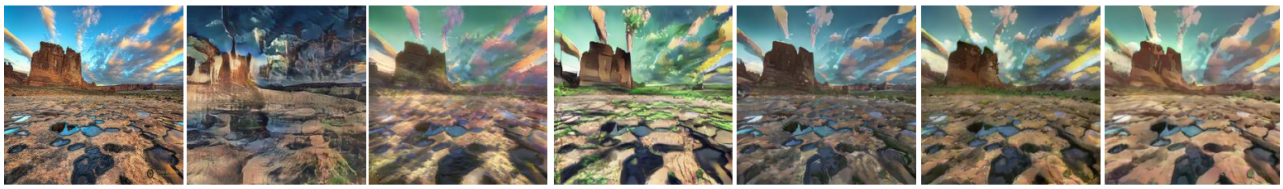


Figure 8: Ablation study of semi-supervised I2I translation. The effect of each key component is illustrated.

Real-Time Video Processing Enhancements:

Work on optimizing the model for faster processing speeds to enable real-time video conversion. This may involve algorithmic improvements and hardware optimization strategies. Explore deploying the model on various platforms, including mobile devices and web applications, to broaden accessibility and user engagement.

User Feedback Integration:

Establish a feedback loop with users to gather insights on the quality and usability of the generated anime videos. This feedback will be critical for iterative improvements and feature development. Consider conducting user studies to assess satisfaction and gather qualitative data on user preferences regarding style fidelity and content retention.

8 Conclusion

In this report, we addressed the challenge of transforming real videos into anime-styled videos while maintaining a balance between style fidelity and semantic accuracy. We used the Scenimefy framework, which effectively facilitates image-to-image translation through three key stages:

Pseudo Paired Data Generation: This stage focuses on creating a dataset that simulates paired data, which is crucial for training the model to understand the relationship between real images and their anime counterparts.

Semantic Segmentation-Guided Data Selection: By leveraging semantic segmentation, we enhanced the data selection process, allowing the model to prioritize relevant features that are vital for achieving a high-quality anime style transformation.

Semi-Supervised Image-to-Image Translation: This approach enables the model to learn from both labeled and unlabeled data, improving its robustness and performance across various input scenarios.

The significance of our results is evident in the seamless anime-style transformations achieved in video format, demonstrating a high degree of texture detail and semantic alignment compared to state-of-the-art baselines.

However, we faced several challenges during our research, including:

High Computation Requirements: The computational demands of the model necessitated access to advanced hardware, impacting the overall efficiency of the training and translation processes.

Detail Preservation Challenges: Ensuring the preservation of fine details in the transformed images was a continual challenge that required careful tuning of the model's parameters.

Semantic Translation Inconsistencies: We observed inconsistencies in semantic translation, where the model sometimes struggled to maintain alignment between the content of the original video and the final anime-styled output.

In conclusion, while the Scenimefy framework represents a significant advancement in the field of anime-style video transformation, further refinement and optimization are needed to address the challenges encountered. Future work will focus on enhancing computational efficiency, improving detail preservation, and mitigating semantic inconsistencies to achieve even more accurate and visually appealing results.

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